CHRIST (DEEMED TO BE UNIVERSITY) • DELHI NCR CAMPUS • SCHOOL OF SCIENCES

EVENT VOLUNTEER HOUR LEDGER

A Relational Database Management System for Service Hour Accounting, Faculty Verification & Certificate Eligibility

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1. Real-World Context & The Spreadsheet Crisis

Institutional Domain: CHRIST University's student wing **AADIKALA** coordinates student leadership across 6 specialized sub-clubs: Dance, Emcee, Technical, Creative, Music, and Hospitality. Festivals like *SUNERGOS* and *TARANG* mobilize hundreds of volunteer service hours.

The 4 Fatal Spreadsheet Flaws:

- **Mutable Update Anomaly:** Spreadsheets store a single mutable 'Total Hours' cell; concurrent edits overwrite historical totals without a paper trail.
- **Multi-Club Join Inflation:** Co-hosted events associate multiple sub-clubs; unnormalized queries multiply volunteer hours by the club count.
- **Premature Certification:** Unreviewed timesheets allow students to falsely claim threshold eligibility before faculty signoff.
- **Zero Forensic Auditability:** Absence of Change Data Capture (CDC) prevents determining who edited, approved, or revoked hours.

2. The Core Solution: The Ledger Paradigm

To permanently eliminate the mutable aggregate anomaly, the schema applies the **Ledger Paradigm** inspired by double-entry financial accounting:

Key Invariant: Total hours are *NEVER* stored as a mutable column in the students table.
Every volunteer interval is an immutable transactional entry in volunteer_hour_entries with start/end timestamps and break duration.

Dynamic Authoritative Aggregation:

```
Approved Hours: H_s = ∑(verified_minutes) / 60.0
WHERE submission_status IN ('approved', 'corrected')
```

Status Isolation: Pending claimed hours and rejected submissions are strictly isolated from certified totals.

3. System Objectives & Policy Governance

Core University Milestone Rule: Students achieving ≥ 20 verified volunteer hours (1,200 minutes) across university events earn an official *Certificate of Volunteer Contribution*.

- Deliver a 3NF/BCNF relational database in MySQL 8.0.
- Enforce engine-level Check Constraints and Foreign Key Actions.
- Implement database triggers for workflow rules and audit logging.
- Wrap certificate issuance in ACID transactions with row-level locks.
- Deploy a public verification engine validating cryptographic tokens.

4. Three-Tier Architecture & ER Design

1. Presentation Layer: React 18 + Vite (TypeScript, Tailwind CSS, Recharts) hosted live on GitHub Pages.
2. Application Layer: Python 3.14 Flask REST API with Role-Based Access Control (RBAC).
3. Database Layer: MySQL 8.0 InnoDB Engine with 10 Normalized Tables:

Relation	Key Attributes	Cardinality & Purpose
users	user_id (PK), email, role	1:1 with students; login identity & RBAC.
students	student_id (PK), user_id (FK)	Academic registry; reg number & programme.
clubs	club_id (PK), parent_club_id	1:N Recursive Self-Join (AADIKALA & sub-clubs).
events	event_id (PK), event_code	1:N with participation; campus event logs.
event_subclubs	(event_id, club_id) (PK)	M:N junction linking events to collaborating clubs.
event_volunteers	participation_id (PK)	M:N student-event associative entity; UNIQUE.
hour_entries	hour_entry_id (PK)	1:N transactional ledger of work intervals.
policies	policy_id (PK)	1:N threshold policies (1,200 min = 20 hr).
certificates	certificate_id (PK)	1:N immutable award snapshot + SHA-256 token.
audit_logs	audit_id (PK)	Change Data Capture before/after JSON states.

5. Normalization Journey & Solving Join Inflation

Decomposition Steps (1NF → 3NF/BCNF):

- **1NF (Atomicity):** Decomposed repeating comma-separated sub-club lists into junction entity event_subclubs(event_id, club_id).
- **2NF (Full Functional Dependency):** Decoupled club attributes (name) and event attributes (dates, venues) from junction composite keys.
- **3NF & BCNF (No Transitive Dependencies):** Isolated authentication from academic enrollment. Removed mutable calculated totals.

The Join Inflation Resolution:

When an event involves 3 sub-clubs, joining clubs directly to hour entries triples student hours! Our design attaches hours strictly to event_volunteers(participation_id). Club summaries use independent COUNT(DISTINCT) aggregations, guaranteeing zero double-counting.

6. Active Triggers & ACID Concurrency

Database Triggers (Workflow & Audit Logging):

- trg_vhe_before_insert: Validates positive claimed minutes & non-negative breaks.
- trg_vhe_before_update: Gatekeeper—rejecting hours without rejection_reason raises SIGNAL SQLSTATE '45000'.
- trg_vhe_after_update: Automatically logs old & new states into audit_logs as JSON.

ACID Certificate Issuance (Row-Level Locking):

In sp_issue_certificate, race conditions are blocked via row locks:

```
START TRANSACTION;
SELECT * FROM policies WHERE policy_id = 1 FOR SHARE;
SELECT * FROM students WHERE student_id = 4 FOR UPDATE;
IF approved_minutes < 1200 THEN ROLLBACK; END IF;
IF active_cert_count > 0 THEN ROLLBACK; END IF;
INSERT INTO certificates (...) VALUES (...);
COMMIT;
```

7. Balanced B-Tree Indexing & Analytics

Indexing Architecture:

- **Primary Clustered:** O(log N) row access across all 10 tables.
- **Secondary B-Tree:** idx_users_email, idx_students_reg_no, idx_cert_token.
- **Composite B-Tree:** idx_vhe_part_status (participation_id, submission_status) speeds view joins.
- **Query Execution (EXPLAIN):** Evaluates joins using index seeks (ref/range) under 50ms, eliminating full table scans (ALL).

6 Analytical Reporting Queries (reports.sql):

1. Eligible Students (≥ 20 Hrs) • 2. Near-Threshold Volunteers (15–19.99 Hrs)
3. Event Labor without Duplication • 4. Sub-Club Breakdown (Dance, Emcee, etc.)
5. Pending Timesheets Impact Simulation • 6. Certificate Traceability Log

8. Production Results & Live Verification

Automated Test Scenarios Verified:

• **Student A (18h):** Ineligible • **Student B (20h):** Eligible for Certificate
• **Student C (18h + 5h pending):** Review Pending • **Student D (24h):** Certified

Live Deployment & Public Token Verifier:

Every certificate is embedded with a SHA-256 token verifiable at the live URL:
https://radhika2405-max.github.io/event-volunteer-hour-ledger/

Conclusion: Successfully models enterprise relational theory to solve real-world university service governance without data anomalies.